

Grade 7
Unit 6
Lesson 9 Practice

Name _____
Date _____
Period _____

For questions 1-2, use the following scenario:

Some wooden chests are in a store. A green chest has a length of 165 centimeters (cm), a width of 70 cm, and a height of 10 cm.

1. If another box, a blue box, is proportional in size, what is its length if its width is 18 cm?

$$\frac{70}{165} = \frac{18}{x}$$

42.4 cm

2. If another box, a red box, is proportional in size, what is its height if its length is 14 cm?

$$\frac{x}{14} = \frac{10}{165}$$

0.85 cm

3. A blue marble can make 8.45 laps around a track in 22.15 seconds (s). Meanwhile, a yellow marble makes 7.50 laps around a track every 15 s. What is the unit rate for the marble that circles the track faster?

yellow $\frac{7.5}{15} = 0.5$ laps per second

blue $\frac{8.45}{22.15} = 0.38$ laps per second

For questions 4-10, use the following scenario:

For a family dinner, AnnaMarie is making a turkey casserole. The recipe serves 8 and calls for the following:

1 lb. egg noodles

6 tbsp. butter

2 cloves garlic

1 lb. cremini mushrooms

$\frac{1}{4}$ cup all-purpose flour

$2\frac{1}{2}$ c. low-sodium chicken broth

1 c. heavy cream

2 lbs. leftover turkey

1 c. frozen peas

If AnnaMarie adjusts the recipe to serve 6 people, she will need:

4. 0.75 lbs. of egg noodles.

$$\frac{1 \text{ lb}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

5. 4.5 tbsp. butter

$$\frac{6 \text{ tbs}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

6. 3 tbsp. of all-purpose flour

$$1 \text{ cup} = 16 \text{ tbsp}$$

$$\frac{1}{4} \text{ cup} = 4 \text{ tbsp}$$

$$\frac{4 \text{ tbsp}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

7. 24 oz. turkey

$$1 \text{ lb} = 16 \text{ oz}$$

$$2 \text{ lb} = 32 \text{ oz}$$

$$\frac{32 \text{ oz}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

8. 12 oz. cremini mushrooms

$$\frac{16 \text{ oz}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

9. 0.75 c. frozen peas

$$\frac{1 \text{ c}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$

10. 1.875 or $1\frac{7}{8}$ c. low-sodium chicken broth

$$\frac{2.5 \text{ c}}{8 \text{ ppl}} = \frac{x}{6 \text{ ppl}}$$